Write and Evaluate Numerical Expressions with Exponents

Evaluate expressions with grouping symbols, powers and arithmetic in one order.

Name _____ **Date** _____

Tonight's spark

A rec-center game gives you 4 points to start. Then you play 3 bonus rounds. Each bonus round is worth $(9 - 7)^2$ points. What is your final score?

Make a guess first: is the final score more or less than 10?

The order, top to bottom

P	Parentheses ( )	1
E	Exponents (powers)	2
M D	Multiply × or divide ÷	3
A S	Add + or subtract −	4

Our steps

Start	$4 + 3 \times (9 - 7)^2$
1. P	$4 + 3 \times 2^2$
2. E	$4 + 3 \times 4$
3. M D	$4 + 12$
4. A S	16

Our class model: rewrite after every step

Try it

Work it out on paper. Write one answer on each line.

1 **In 5 × (2 + 4), which do you do first?**

A. Multiply 5 × 2 **B.** Multiply 5 × 4 **C.** Add 5 + 2 **D.** Add 2 + 4

My answer _____

2 **What is (8 – 3) × 2?**

A. 2 **B.** 10 **C.** 13

My answer _____

3 **What is $(6 + 2)^2$?**

A. 64 **B.** 40 **C.** 10 **D.** 16

My answer _____

4 **What is $20 – (3 + 1)^2$?**

A. 18 **B.** 16 **C.** 4 **D.** 256

My answer _____

5 **What is 2 + 3 × (5 – 1)?**

A. 20 **B.** 14 **C.** 16

My answer _____

Talk about it

What did you do first, second and third, and why does that order give only one answer?

Home mission

Make a game score at the table. Start with 6 points, then add 2 rounds worth $(6 – 3)^2$ points each. Ask your student to say the expression and rewrite it after every step.

If we get stuck, ask

- What step comes first, and why?
- Can you rewrite the whole expression after that step?
- These two operations have the same rank — which way do we go?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/6-4-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it